

Annealed tiling for HSC + MBQ1 on the HSC-Niji wide footprint

D. Schlegel, 2026-10-08. Code: `skytiling (optimize_tiling_maps)`, data in `data/subaru/`.

Setup. The camera is described by per-sub-filter maps of relative throughput on the focal plane, built from Hironao's MBQ1 dome flats (gain-corrected, sky-area-per-pixel and dome-illumination gradient divided out, ≈ 1 over the bulk of the field, 0 where the N0_DATA mask flags the $\sim 10.4'$ -wide filter-holder cross, the vignetted edge beyond $46.5'$, dead amplifiers and the two CCDs absent from the 413 flats). Every pointing is observed at four instrument rotations 90° apart, and the coverage of a sky point in sub-filter f is the sum of throughput over all exposures. Starting from a Fibonacci lattice of **3207 pointings** inside the 1194 deg^2 footprint (0.75° discs around the 802 HSC-SSP Wide centres at $-6^\circ < \text{Dec} < 6.4^\circ$) — the same number of exposures as Hironao's phase-1 pattern (4 dithers $\times$ 802 pointings $\times$ 4 PAs) — simulated annealing moves each pointing centre ($\leq 0.75^\circ$) and the rotation offset of its 4-PA set to minimise the summed per-filter coverage variance over 4M random points. 277 iterations on one Perlmuter node (~ 6 s per iteration with 64 workers; step size $0.1^\circ \rightarrow 0.03^\circ$, $10^\circ \rightarrow 3^\circ$ at iteration 77).

Result (footprint interior, 68% of the area, away from the stripe edges; coverage = effective number of full-throughput exposures):

	413	439	465	490
Hironao phase 1: mean $\pm$ rms (rms/mean)	2.85 ± 1.19 (0.42)	3.24 ± 1.12 (0.34)	3.24 ± 1.16 (0.36)	3.11 ± 1.10 (0.35)
fraction $< 1.5 / < 0.5$	13.0% / 2.1%	6.5% / 0.8%	7.4% / 1.2%	7.7% / 0.9%
Fibonacci lattice, un-annealed	2.79 ± 1.08 (0.39)	3.19 ± 1.08 (0.34)	3.17 ± 1.15 (0.36)	3.06 ± 1.04 (0.34)
Annealed (positions + rotations)	2.56 ± 0.86 (0.34)	2.94 ± 0.88 (0.30)	2.91 ± 0.85 (0.29)	2.81 ± 0.84 (0.30)
fraction $< 1.5 / < 0.5$	10.2% / 0.8%	4.9% / 0.3%	4.7% / 0.2%	5.8% / 0.3%

For the same number of exposures the annealed tiling lowers the rms of the coverage by 24–27% in every filter (rms/mean by 13–19%), cuts the deep holes (coverage < 0.5) by a factor 2.8–5, and removes the periodic pattern of Hironao's layout (Figure 1). The mean is $\sim 10\%$ lower because pointings near the stripe boundary move outward to flatten the edge and the coverage they put outside the footprint is not counted; in the interior the annealed solution is more uniform at every quantile. 413 remains the worst filter in both cases because of its two missing CCDs. The 0/1 (GOOD) and throughput-weighted statistics differ by only 2–4%.

Rotations. Allowing the 4-PA set of each pointing to turn is worth a few per cent in rms on top of the position moves. The annealed offsets (Figure 2, lower left) have an rms of 17° , with 57% of pointings turned by more than 10° and 9% by more than 30° ; the distribution is broad and single-peaked with a small positive mean ($+3.5^\circ$), i.e. the solution does not favour a particular grid of angles. The pointings themselves moved by a median 0.19° from the lattice (max 0.75° = the cap, reached by only 3 pointings), so the Fibonacci lattice is a good starting point and the result is set by the camera masks and the tile density rather than by the start.

Convergence. The objective fell from 1.58 to 1.24 (sum over filters of rms/mean); 85% of the gain came in the first 40 iterations, and iterations 207–276 gained only 0.005 (Figure 2, upper left). At the final slope of $6e-5$ per iteration, another 300 iterations would buy well under 1%, so the run is converged for practical purposes at this tile density. Larger gains would come from more tiles (lower rms/mean at higher mean coverage) or from a different objective (e.g. penalising holes rather than variance).

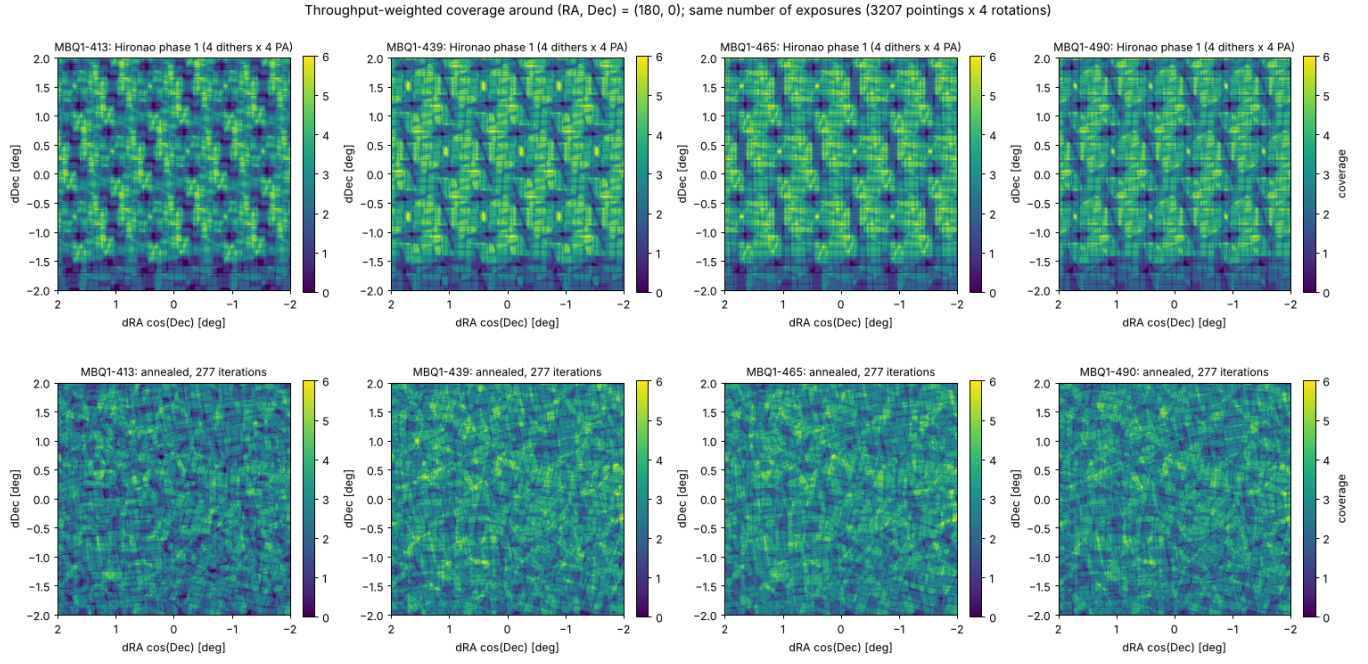


Figure 1. Throughput-weighted coverage per sub-filter in a $4^\circ \times 4^\circ$ field around (RA, Dec) = (180°, 0°) for Hironao's phase-1 pattern (top) and the annealed tiling (bottom, iteration 276), same colour scale.

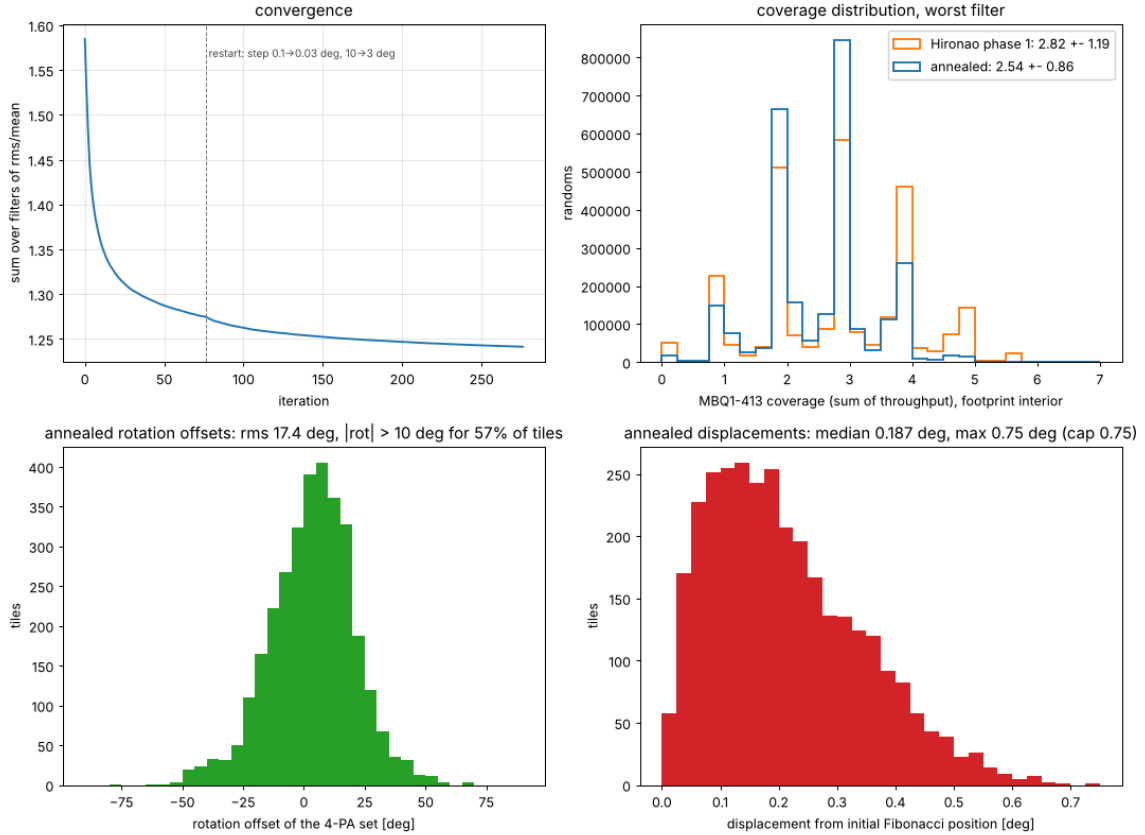


Figure 2. Convergence of the objective; distribution of MBQ1-413 coverage over the footprint interior; annealed rotation offsets; and displacement of each pointing from its initial lattice position.

Files: hsc_fib/fib_tiles_0276.fits (columns ra, dec, rot, ra0, dec0; each row is one pointing to be observed at rot, rot+90, rot+180, rot+270), hsc_fib/fib_randoms_4000000.fits. The mapping of rot onto INSR0T_PA (sign and zero point) still has to be calibrated against the instrument.